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$$\Delta U_{ab} = Q_{a \rightarrow b} - W_{a \rightarrow b} = 100 \text{ kJ} - 40 \text{ kJ} = \underline{60 \text{ kJ}}$$

$$\Delta U_{ab} = Q_{a \rightarrow b} - W_{a \rightarrow b}$$

$$Q_{a \rightarrow b} = \Delta U_{ab} + W_{a \rightarrow b} = 60 \text{ kJ} + 20 \text{ kJ} = \underline{80 \text{ kJ}}$$

$$Q_{b \rightarrow a} - W_{b \rightarrow a} = \Delta U_{ba}$$

$$Q_{b \rightarrow a} = \Delta U_{ba} + W_{b \rightarrow a}$$

$$= -60 \text{ kJ} - 30 \text{ kJ} = \underline{-90 \text{ kJ}}$$

$$U_a = 0 \text{ kJ} \quad U_d = 40 \text{ kJ}$$

$$\therefore \Delta U_{ab} = 60 \text{ kJ} \quad \therefore U_b = 60 \text{ kJ}$$

由 d-b 为定容过程, 系统不对外做功,

$$\text{故 } W_{a \rightarrow d} = W_{a \rightarrow b} = 20 \text{ kJ}$$

$$a \rightarrow b \text{ 过程与外界交换的热量 } Q_{ab} = U_b - U_a = 60 \text{ kJ} - 40 \text{ kJ} = \underline{20 \text{ kJ}}$$

$$Q_{a \rightarrow d} = \Delta U_{ad} + W_{a \rightarrow d} = 40 \text{ kJ} - 0 \text{ kJ} + 20 \text{ kJ} \\ = \underline{60 \text{ kJ}}$$

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过程	Q/kJ	W/kJ	$\Delta U/\text{kJ}$
1-2	1390	0	1390
2-3	0	395	-395
3-4	-1000	0	-1000
4-1	0	-5	5

2-7

选取气缸内的空气为研究的热力学系统，

系统的初始压强为：

$$760 \text{ mmHg} \approx 1.013 \times 10^5 \text{ Pa}$$

$$P_1 = P_{\text{atm}} + \frac{195 \times 9.81}{100 \times 10^{-4}}$$

$$771 \text{ mmHg} \approx 1.028 \times 10^5 \text{ Pa}$$

$$= 1.028 \times 10^5 + 1.91295 \times 10^5 \text{ Pa}$$

$$= 2.94025 \times 10^5 \text{ Pa}$$

减少负载后，系统压强为：

$$P_2 = P_{\text{atm}} + \frac{95 \times 9.81}{100 \times 10^{-4}}$$

$$= 1.028 \times 10^5 + 0.93195 \times 10^5 \text{ Pa}$$

$$= 1.95995 \times 10^5 \text{ Pa} \approx 1.960 \times 10^5 \text{ Pa}$$

由题可知，气体与外界充分换热，可将过程看成等温过程

又由理想气体的状态方程可得：

$$Pv = \text{const}$$

$$P_1 V_1 = P_2 V_2$$

$$V_2 = \frac{P_1 V_1}{P_2} = \frac{2.940 \times 10^5 \times 100 \times 10^{-4} \times 10 \times 10^{-2}}{1.960 \times 10^5}$$

$$= 1.5 \times 10^{-3} \text{ m}^3$$

$$\text{活塞上升的距离 } \Delta h = \frac{V_2 - V_1}{A} = \frac{1.5 \times 10^{-3} - 100 \times 10 \times 10^{-6}}{100 \times 10^{-4}}$$

$$= 0.05 \text{ m} = 5 \text{ cm}$$

系统的初始温度与终止温度相等 $\Rightarrow$ 内能变化为 0

可看作定压膨胀过程

$$Q = W = p_2 \Delta V = p_2 A \Delta h = 1.960 \times 10^5 \times 100 \times 10^{-4} \times 0.08 = \underline{98 \text{ J}}$$

2-8

$$(1) W = q - \Delta u = -50 \text{ kJ/kg} - 146.5 \text{ kJ/kg} = \underline{-196.5 \text{ kJ/kg}}$$

$$(2) \Delta h = \Delta(u + pV)$$

$$W_s = q - \Delta h = -50 - 146.5 - (0.8 \times 10^{-3} \times 0.175 - 0.1 \times 10^{-3} \times 0.845) = \underline{-252.6 \text{ kJ/kg}}$$

$$(3) P = -\frac{W_s \times 10}{60} = \underline{42 \text{ kW}}$$

2-11

此过程为开口系的稳流过程

$$Q = h_6 q_{m6} - h_7 q_{m7} - h_1 q_{m1} + W_s$$

$$q_{m6} = q_{m7} + q_{m1}$$

$$Q = h_6 (q_{m7} + q_{m1}) - h_7 q_{m7} - h_1 q_{m1} + W_s$$

$$= q_{m7} (h_6 - h_7) + q_{m1} (h_6 - h_1) + W_s$$

$$P = W_s = Q - q_{m7} (h_6 - h_7) - q_{m1} (h_6 - h_1)$$

$$= \frac{1}{3600} [700 \times 418200 - 700(418 - 42) - 50 \times 10^3 (418 - 12)]$$

$$= 2415.7778 \text{ kW}$$